Pipe, Inspect, Extract: (PIE)

User Manual

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Version: 3.2.x

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Introduction

1.1 Overview

Pipe, Inspect, Extract (PIE) is a data visualization system designed for analyzing pipeline data to detect defects using various plots and heatmaps.

1.2 Purpose

The purpose of this software is to provide fast and reliable pipeline defect analysis:

- User Interface: Centralized view of pipe data, defects and dig sheet.
- Visualization: Heatmaps, line charts, 3D views, and proximity line charts.
- Reporting: Generate preliminary, final and dig sheet reports.
- Analysis: Anomaly distribution, XYZ mapping, pipe highlight.

1.3 Data

The client will be given pipe wise data in a specific folder. Now the user has to load this folder from the app. *(Fig. 1-1)*

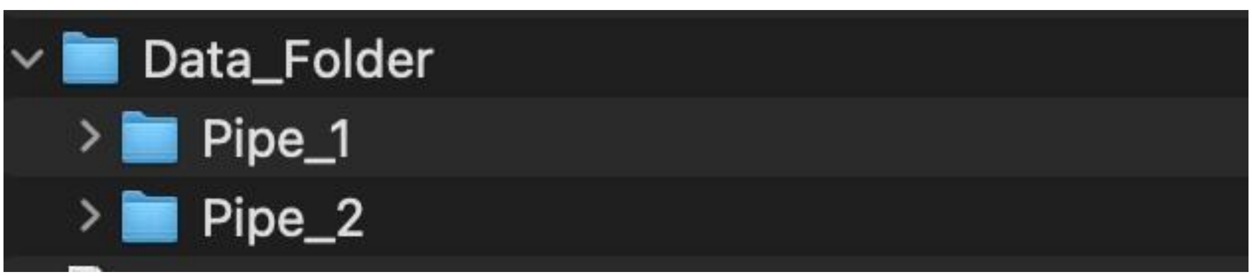

Fig-1.3

Note: Each Pipe folder contains various file formats, so it is advised to not alter or modify any of its contents.

Installation and Setup

2.1 System Requirements

Minimum RAM: 8GB

Minimum Processor: i5 and (or equivalent) 5th generation

Operating System: Windows 10/11

Disk Space: Minimum of 2GB of free disk space for installation

Graphics Card: DirectX 11 compatible graphics card with at least 512MB of VRAM (for optimal performance)

2.2 Installation Instructions

Windows: Click the downloaded .exe file (e.g., *PIE.exe*) to start the installation process. If prompted by User Account Control (UAC), click "Yes" to allow the installer to make changes to your system.

Getting Started

3.1 Initial Setup

When the app loads, we first need to load the project (Data folder of pipes).

You can do that by clicking on: ***File > Create Project***

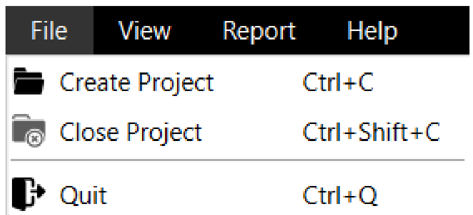

Fig-3.1

3.2 First-Time Use

After creating a project, use the Pipe dropdown and Load button to open a pipe. Tabs and visualizations update dynamically when switching between views.

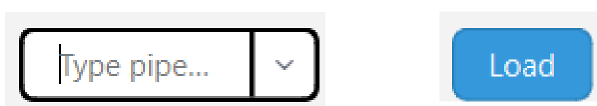

Fig-3.2

User Interface

4.1 Main Components

Some components of software are:

Menu Bar

Menu Bar is used to trigger action buttons for various functions of the software and provides access to project management, analysis, reporting and help.

Fig-4.1

4.2 Navigation

We can navigate through different tabs of the Tab Widget, which are:

1) Defect Sheet Tab

Here we can see the calculated defects and all their attributes like; length, width, depth peak, orientation, feature class, etc.

	Defect_id	Abs. Distance (m)	Distance to U/S GW(m)	Pipe Number	Pipe Length (mm)	Feature Identification	Dimensions Classificatior	Orientation o' clock	Length (mm)	Width (mm)
1	64	620.196	0.843	35	12.51	Corrosion	PITTING	08:37:39	35	18
2	65	622.123	2.193	35	12.51	Corrosion	AXIAL GROOVING	09:14:54	49	22
3	66	624.09	3.55	35	12.51	Corrosion	PITTING	09:22:32	28	18
4	67	625.7	7.401	35	12.51	Corrosion	PITTING	09:23:01	31	22
5	68	627.219	10.199	35	12.51	Corrosion	GENERAL	08:24:21	115	47
6	69	628.19	10.706	35	12.51	Corrosion	AXIAL GROOVING	00:23:22	49	18
7	70	629	11.273	35	12.51	Corrosion	GENERAL	07:37:28	210	55

Fig-4.2.1

2) Tab Switcher Dropdown

Here the user can import different types of plots from the data.

Fig-4.2.2

3) Toggle Buttons

- Show/Hide Table: switches defect table visibility.
- Stack/Side-by-Side: switches heatmap layout vertically or horizontally.

Fig-4.2.3

Features

5.1 Anomaly Distributions - Graphs

Generate distribution plots of anomalies such as Metall Loss defects, ERF, Orientation and so on.

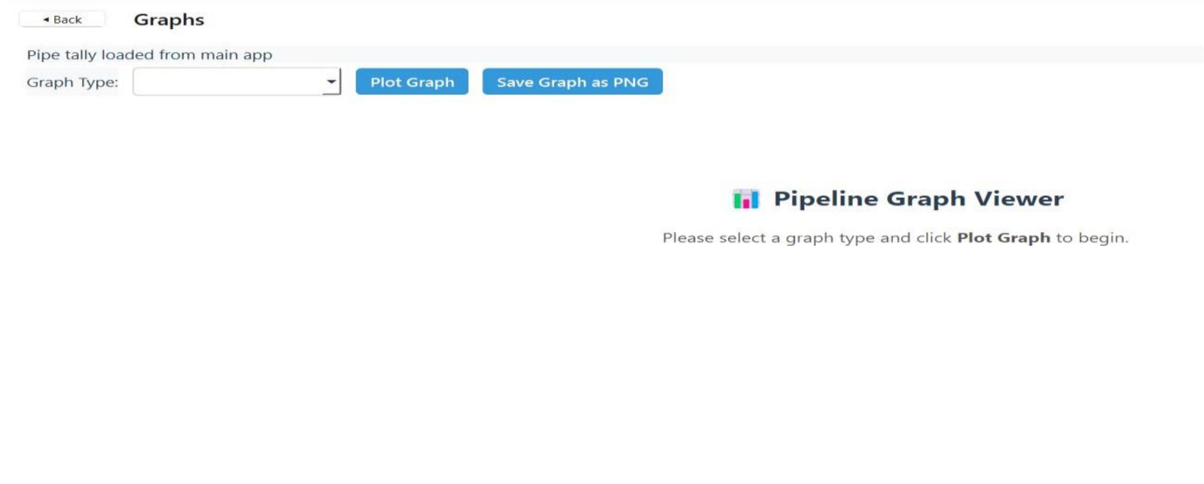

Fig-5.1

5.2 Heatmap Defects

Color-coded defect visualizations with ability to automatically detect and highlight major defects in heatmap

Both heatmaps show corresponding regions of the pipeline for easy comparison of data sources (e.g., Hall sensor vs. Proximity sensor).

Users can switch layout using the **“Stack / Side-by-Side”** button.

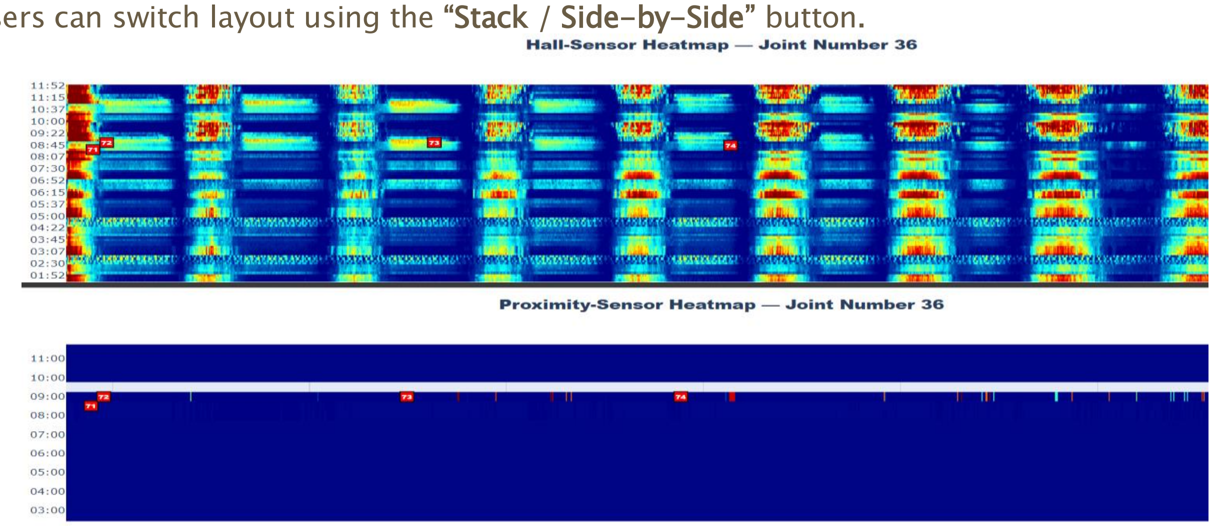

Fig-5.2

5.3 Line Charts

Line charts on basis of:

- Proximity Sensor
- Hall Sensor

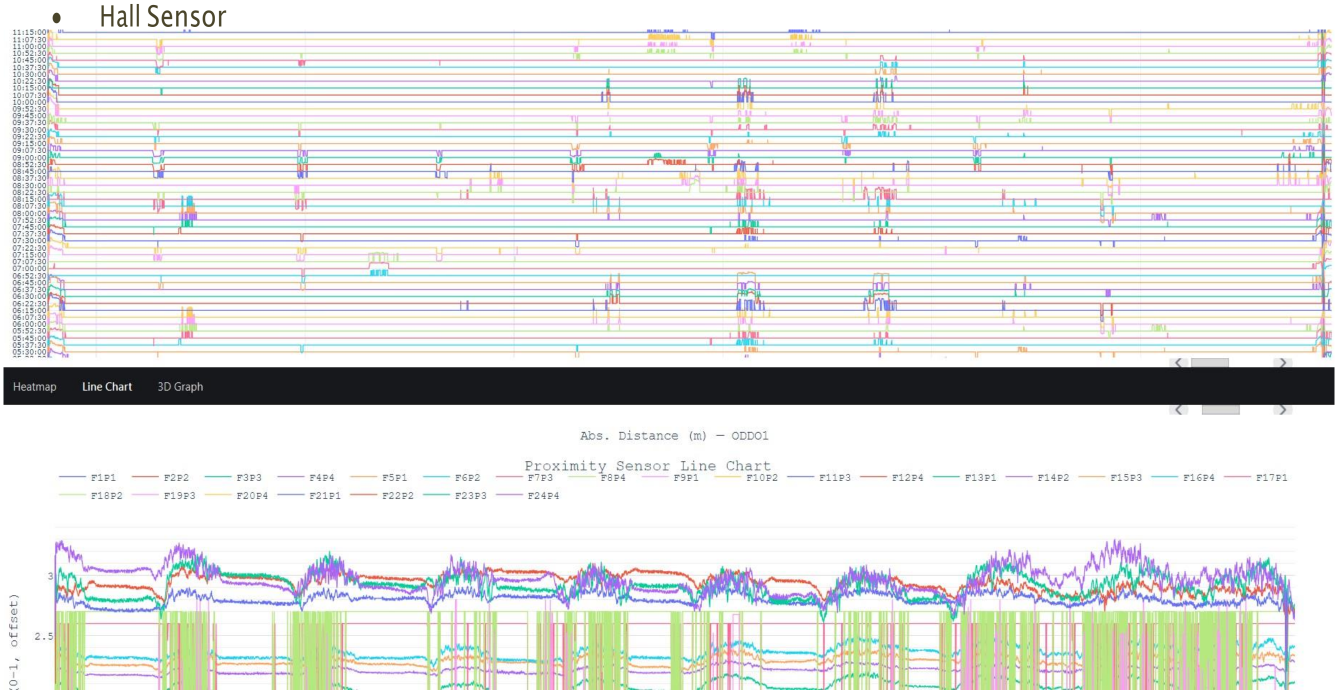

Fig-5.3

5.4 XYZ Mapper

XYZ Mapper will be used to integrate KML maps, enabling users to visualize data pipelines geographically, providing a spatial context for analyzing pipeline conditions in relation to their physical locations.

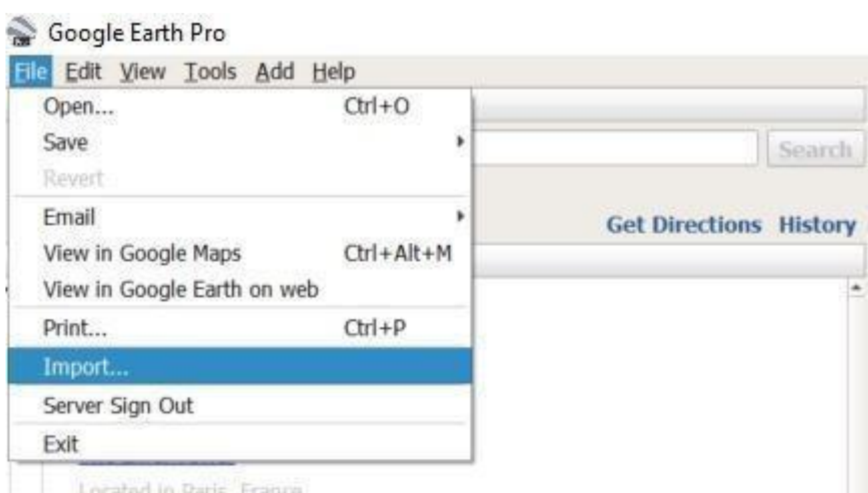

Fig-5.4

5.5 Pipe Highlight

This window displays all the statistics of anomalies related to pipeline in visualizations. The data is taken from Pipe Tally.

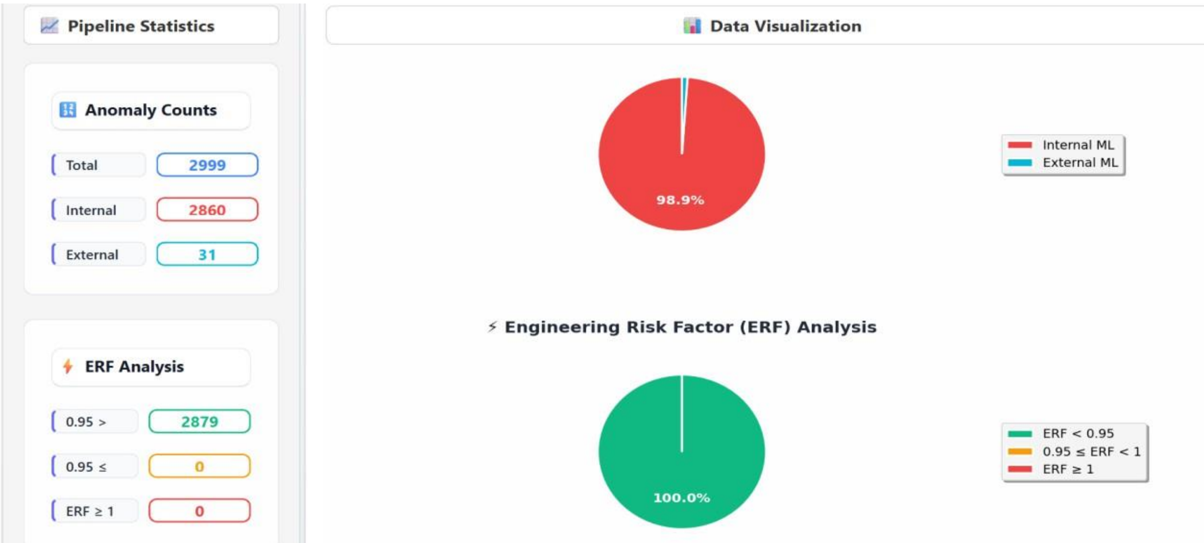

Fig-5.5

5.6 Pipe Scheme

The Pipe Scheme feature provides a comprehensive schematic representation of each pipeline segment, showing pipe connections, weld joints, and identified anomalies in a simplified engineering layout.

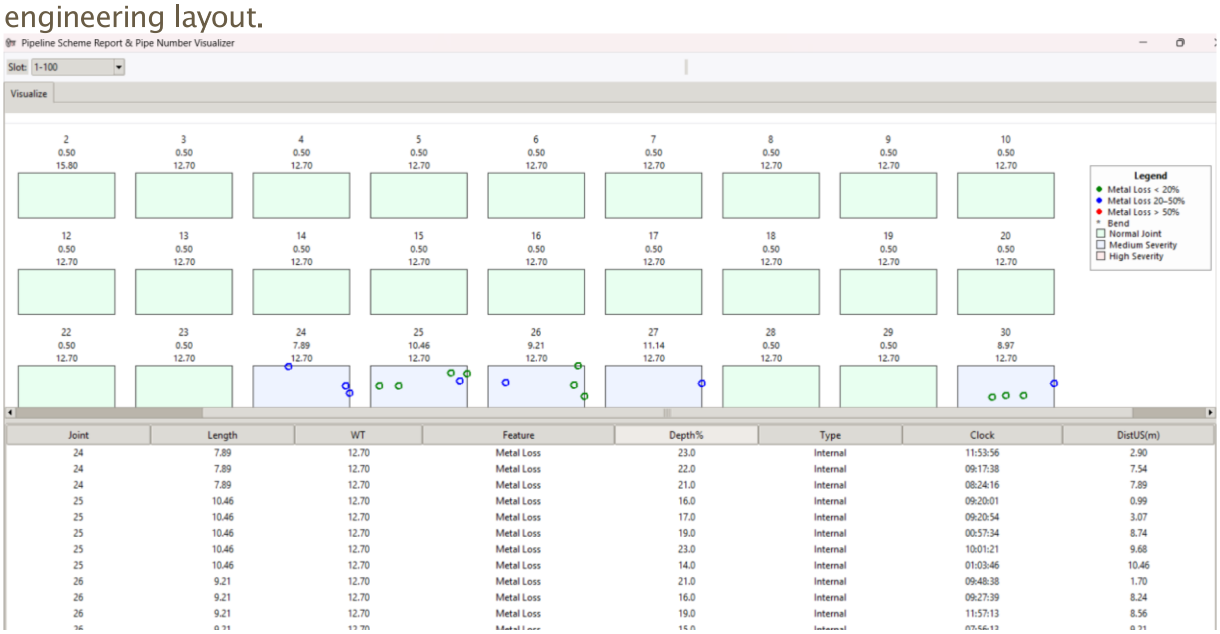

Fig-5.6

5.7 Reports

In this, it generates Preliminary and Final reports.

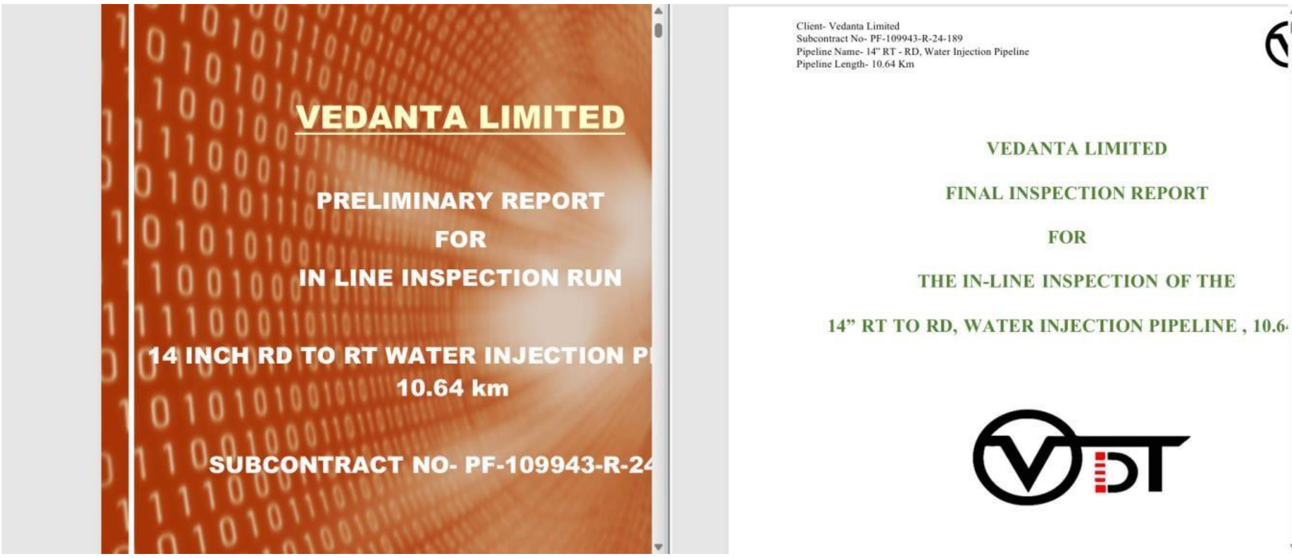

Fig-5.7

Dig Sheet

6.1 Introduction

Users can use this inbuilt software to visualize and generate calibrated reports of various defects found by our software. It is a proprietary software developed with PIE.

6.2 Use

- Standard Dig Sheet: Access via Report > Generate > Dig Sheet.
- Dig Sheet by Absolute Distance: Enabled when selecting a defect row in the Defect Table.

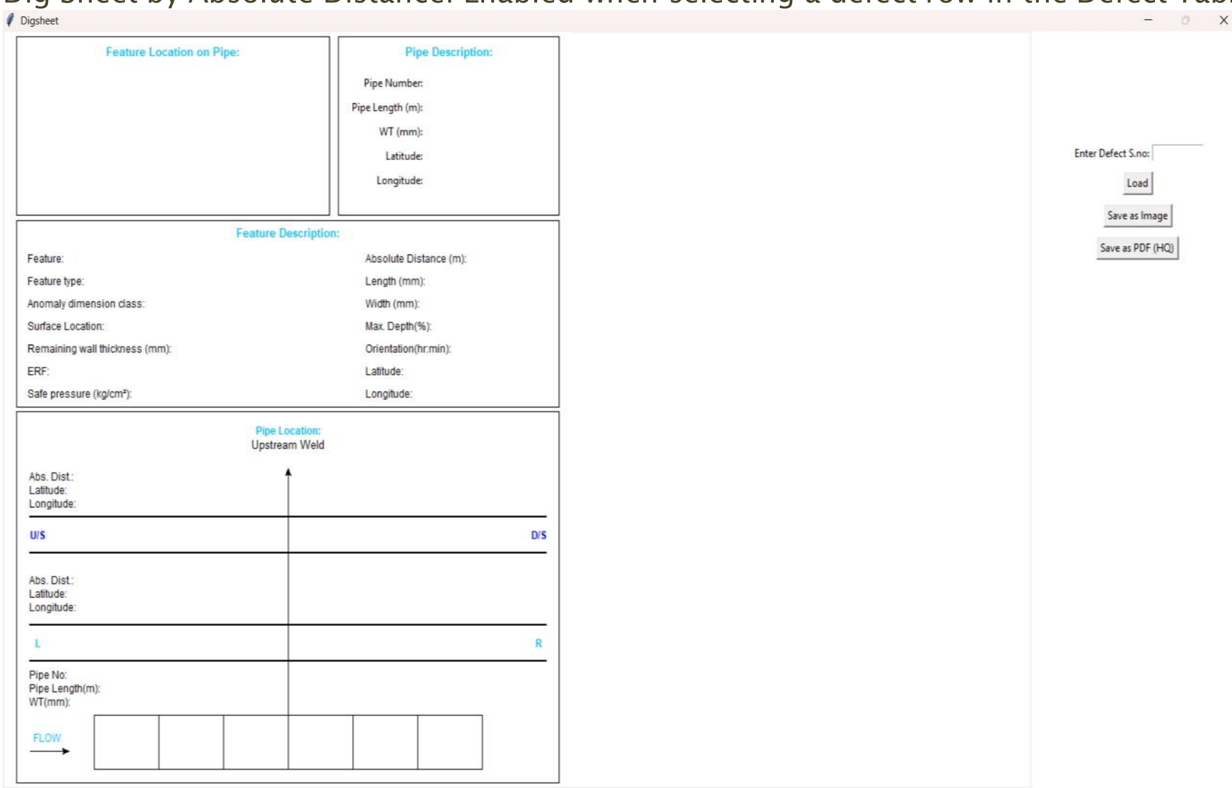

Fig-6.2

Appendices

7.1 Glossary of Terms

Pipe, Inspect, Extract (PIE)

A data visualization system designed for analyzing pipeline data to detect defects using various plots and heatmaps.

Visualization

The process of generating graphical representations of data to aid in analysis, such as custom plots, telemetry plots, anomalies distribution plots, and heatmaps.

Telemetry Plotting

The method of displaying sensor data over time to observe changes and trends in the pipeline data.

Defect Boxing

The feature that allows users to magnify and identify defects within a heatmap for detailed inspection.

Custom Plotting

The ability to create and view various types of plots based on user-defined parameters and data selections.

Heatmap

A data visualization tool that uses color coding to represent the magnitude of values in a matrix or grid, highlighting areas of interest, such as defects in pipeline data.

Tab Widget

A user interface component that organizes different views or sections within the software, such as Defect Sheet, Data Tab, and Plot Tab.

Back Screen

The area of the interface that displays plots and heatmaps in a minimized format when switching between tabs.

Magnetisation View

A plot that shows the relationship between the magnetisation values and the odometer readings, which helps in understanding the magnetic properties of the pipeline over its length.

Velocity View

A plot that represents the velocity of tool sensors, helping to analyze the speed and consistency of data collection.

Defect ID

A unique identifier assigned to each detected defect, used to retrieve detailed information about the defect from the dig sheet report.

Data Tab

A section within the software that displays raw sensor data collected from the pipeline, allowing for in-depth analysis.

Plot Tab

A section where users can create and view different types of plots based on the data collected from the pipeline.

Dig Sheet

A report generated by the software that provides detailed, calibrated information about detected defects, including attributes like length, width, and orientation.

Status Bar

A component of the user interface that shows the current status of the application and the time of data loading.

Tool Bar

A component of the user interface that contains tools for processing and exporting defect sheets, among other functions.

Menu Bar

A component of the user interface that provides access to various action buttons and functions of the software.

Magnetic Flux Leakage (MFL)

A non-destructive testing method used to detect defects in ferromagnetic materials, such as pipelines, by measuring changes in magnetic flux.